



Programme Module for

Computational Methods and Problem Solving

leading to

Level 5 QQI

Computational Methods and Problem Solving 5N0554

Introduction

This programme module may be delivered as a standalone module leading to certification in a QQI minor award. It may also be delivered as part of an overall validated programme leading to a Level 5 QQI Certificate.

The teacher/tutor should familiarise themselves with the information contained in Dublin and Dún Laoghaire ETB's programme descriptor for the relevant validated programme prior to delivering this programme module.

The programme module is structured as follows:

1. Title of Programme Module
2. QQI Component Title and Code
3. Duration in hours
4. Credit Value of QQI Component
5. Status
6. Special Requirements
7. Aim of the Programme Module
8. Objectives of the Programme Module
9. Learning Outcomes
10. Indicative Content
11. Assessment a. Assessment Technique(s) b. Mapping of Learning Outcomes to Assessment Technique(s) c. Guidelines for Assessment Activities
12. Grading
13. Learner Marking Sheet(s), including Assessment Criteria

Integrated Delivery and Assessment

The teacher/tutor is encouraged to integrate the delivery of content where an overlap between content of this programme module and one or more other programme modules is identified. This programme module will facilitate the learner to develop the academic and vocational language, literacy and numeracy skills relevant to the themes and content of the module.

Likewise the teacher/tutor is encouraged to integrate assessment where there is an opportunity to facilitate a learner to produce one piece of assessment evidence which demonstrates the learning outcomes from more than one programme module. The integration of the delivery and assessment of level 5 Communications and level 5 Mathematics modules with that of other level 5 modules is specifically encouraged, as appropriate.

Indicative Content

The indicative content in Section 10 does not cover all teaching possibilities. The teacher/tutor is encouraged to be creative in devising and implementing other approaches, as appropriate. The use of examples is there to provide suggestions. The teacher/tutor is free to use other examples, as appropriate. The indicative content ensures all learning outcomes are addressed but it may not follow the same sequence as that in which the learning outcomes are listed in Section 9. It is the teacher's/tutor's responsibility to ensure that all learning outcomes are included in the delivery of this programme module.

1. Title of Programme Module
Computational Methods and Problem Solving
2. Component Name and Code
Computational Methods and Problem Solving 5N0554
3. Duration in Hours
150 Hours (typical learner effort, to include both directed and self directed learning)
4. Credit Value
15 Credits
5. Status
This programme module may be compulsory or optional within the context of the validated programme. Please refer to the relevant programme descriptor, Section 9 Programme Structure
6. Special Requirements
None.
7. Aim of the Programme Module
This programme module aims to equip the learner with the knowledge, skill and competence to apply a broad range of computational methods and problem skills and tools to a wide variety of contexts, with some theoretical understanding.
8. Objectives of the Programme Module
• To facilitate the learner in their progression of using computational methods and problem solving for working and life • To assist the learner to develop a greater understanding of computational methods and problem solving through the medium of the indicative content • To facilitate the development of student-centred learning and enable the learner to take responsibility for his/her own learning • To assist the learner to develop skills in the following area of computational methods and problem solving: Discrete Computational Structures and their applications, discrete Probability, Computational modelling and simulation, Linear algebra and applications, Algorithms and Complexity • To equip the learner with the skills and tools that allow them to make appropriate and informed decisions in order to solve problems, automate systems and transform data

9. Learning Outcomes of Level 5 Computational Methods and Problem Solving 5N0554

Learners will be able to:

1. Discrete Computational Structures and their applications

Demonstrate knowledge and understanding of numeric and structural data representations and usages to include:

- Data structures/representations such as arrays, lists, matrices, trees, and their use e.g. ASCII Art, Computer Graphics, Gaussian Elimination (solving linear equations), Decision Matrices, Classical methods for sorting, searching and filtering data

Functions and Recurrence Relations

Demonstrate knowledge and understanding of the salient features of iterative and recursive algorithms and of where to apply each

2. Discrete Probability

Demonstrate knowledge and understanding of elementary probability and information-theory concepts to include:

- Elementary probability distributions (e.g. Gaussian), statistics (Sample Mean and Sample Variance) and their applications
- Calculate probabilities of events and expectations of random variables for elementary problems such as games of chance
- How to Differentiate between dependent and independent events
- Real-world applications of conditional probabilities to problem solving
- The use of random numbers in computing
- Methods for generating pseudo- random numbers
- The significance of randomness to modern computing systems e.g. PKI (Public Key Infrastructure)

3. Computational Modelling and Simulation

Demonstrate knowledge and understanding of basic computer simulation methods to include:

- Computational methods
- Numerical methods e.g. Monte Carlo methods

4. Linear Algebra and Applications

Apply knowledge and understanding of numeric and structural representations and usages to address computational approaches to real-world problems

- e.g.: practically employ use cases, of arrays/matrices e.g. in graphics, games scenarios, in Google PageRank algorithm, sequence alignment or nearest neighbour problems

5. Algorithms and Complexity

Apply knowledge and understanding of Know-how and skill Range Demonstrate a broad range of specialised skills and algorithms to address computational approaches to real-world problems

- e.g.: assess the appropriateness of an algorithm or computational approach, for example in terms of speed, efficiency, best/expected/worst case behaviours

6. Applications of Discrete Probability

Apply knowledge and understanding of elementary probability and information-theory to address computational approaches to real-world problems

- e.g.: Conduct average case analysis of algorithms, perform failure prediction, perform software branch prediction, analyse network packet loss

7. Computational Modelling and Simulation

Differentiate between the principles of modelling and simulation and explain their use of abstraction that allows a machine to address a real-world problem e.g. modelling the temperature in server room, or robot control, or software for an autonomous vehicle

8. Identify approaches to problem definition, solution design, testing and evaluation

9. **Outline the strengths and weaknesses of**, and primary areas of application for, a range of contemporary problem definition and analysis techniques e.g. brute force, divide-and-conquer and heuristic strategies

10. **Distinguish between** pragmatic solving of a problem (using logic to treat the symptoms) and semantics (interpretation of the problem to establish the root cause)

11. **Engage in an iterative process** involving model creation and validation of the correspondence between the model and the real-world situations being modelled e.g.: create a simple, formal mathematical model of a real-world situation and use that model in a simulation e.g. modelling of the temperature in server room, or robot control, or software for an autonomous vehicle, or modelling traffic flow at an intersection

12. **Demonstrate awareness** of the role of personal attributes such as initiative, methodical approach, logical reasoning, persistence and lateral thinking, in prevention and resolution of problems

13. **Reflect on the impact** of employing numerical and logical thinking principles and concepts in the real world, including accuracy, precision, and decisions that may be made based on computer simulations and models

10. Indicative Content This section provides suggestions for programme content but is not intended to be prescriptive. The programme module can be delivered through classroom based learning activities, group discussions, one-to-one tutorials, field trips, case studies, role play and other suitable activities, as appropriate.
Section 1 : Discrete Computational Structures and their applications
The learner will be facilitated to: LO1 Demonstrate knowledge and understanding of numeric and structural data representations and usages to include: • Data structures/representations such as arrays, lists, matrices, trees, and their use e.g. ASCII Art, Computer Graphics, Gaussian Elimination (solving linear equations), Decision Matrices, Classical methods for sorting, searching and filtering data Functions and Recurrence Relations • Demonstrate knowledge and understanding of the salient features of iterative and recursive algorithms and of where to apply each
Section 2: Discrete Probability
The learner will be facilitated to: LO2 Demonstrate knowledge and understanding of elementary probability and information-theory concepts to include: • Elementary probability distributions (e.g. Gaussian), statistics (Sample Mean and Sample Variance) and their applications • Calculate probabilities of events and expectations of random variables for elementary problems such as games of chance • How to Differentiate between dependent and independent events • Real-world applications of conditional probabilities to problem solving • The use of random numbers in computing • Methods for generating pseudo- random numbers • The significance of randomness to modern computing systems e.g. PKI (Public Key Infrastructure)
Section 3: Computational Modelling and Simulation
The learner will be facilitated to: LO3 Demonstrate knowledge and understanding of basic computer simulation methods to include: • Computational methods • Numerical methods e.g. Monte Carlo methods

Section 4: Linear Algebra and Applications
The learner will be facilitated to: LO4 Apply knowledge and understanding of numeric and structural representations and usages to address computational approaches to real-world problems e.g.: practically employ use cases, of arrays/matrices e.g. in graphics, games scenarios, in Google PageRank algorithm, sequence alignment or nearest neighbour problems
Section 5: Algorithms and Complexity
The learner will be facilitated to: LO5 Apply knowledge and understanding of Know-how and skill Range Demonstrate a broad range of specialised skills and algorithms to address computational approaches to real-world problems e.g.: assess the appropriateness of an algorithm or computational approach, for example in terms of speed, efficiency, best/expected/worst case behaviours
Section 6: Applications of Discrete Probability
The learner will be facilitated to: LO6 Apply knowledge and understanding of elementary probability and information-theory to address computational approaches to real-world problems e.g.: Conduct average case analysis of algorithms, perform failure prediction, perform software branch prediction, analyse network packet loss
Section 7: Computational Modelling and Simulation
The learner will be facilitated to: LO7 Differentiate between the principles of modelling and simulation and explain their use of abstraction that allows a machine to address a real-world problem e.g. modelling the temperature in server room, or robot control, or software for an autonomous vehicle
LO8. Identify approaches to problem definition, solution design, testing and evaluation LO9. Outline the strengths and weaknesses of, and primary areas of application for, a range of contemporary problem definition and analysis techniques e.g. brute force, divide-and-conquer and heuristic strategies
LO10. Distinguish between pragmatic solving of a problem (using logic to treat the symptoms) and

semantics (interpretation of the problem to establish the root cause)
LO11. Engage in an iterative process involving model creation and validation of the correspondence between the model and the real-world situations being modelled • e.g.: create a simple, formal mathematical model of a real-world situation and use that model in a simulation e.g. modelling of the temperature in server room, or robot control, or software for an autonomous vehicle, or modelling traffic flow at an intersection
LO12. Demonstrate awareness of the role of personal attributes such as initiative, methodical approach, logical reasoning, persistence and lateral thinking, in prevention and resolution of problems
LO13. Reflect on the impact of employing numerical and logical thinking principles and concepts in the real world, including accuracy, precision, and decisions that may be made based on computer simulations and models

11. Assessment**11a. Assessment Techniques**

Skills Demonstrations (4)	70%
Examination 1.5 hours	30%

11b. Mapping of Learning Outcomes to Assessment Techniques

In order to ensure that the learner is facilitated to demonstrate the achievement of all learning outcomes from the component specification; each learning outcome is mapped to an assessment technique(s). This mapping should not restrict an assessor from taking an integrated approach to assessment.

	Learning Outcome	Assessment Technique
LO1	Demonstrate knowledge and understanding of numeric and structural data representations and usages to include: Data structures/representations such as arrays, lists, matrices, trees, and their use e.g. ASCII Art, Computer Graphics, Gaussian Elimination (solving linear equations), Decision Matrices, Classical methods for sorting, searching and filtering data Demonstrate knowledge and understanding of the salient features of iterative and recursive algorithms and of where to apply each	Exam/Skills Demo
LO2	Demonstrate knowledge and understanding of elementary probability and information-theory concepts to include: - Elementary probability distributions (e.g. Gaussian), statistics (Sample Mean and Sample Variance) and their applications - Calculate probabilities of events and expectations of random variables for elementary problems such as games of chance - How to Differentiate between dependent and independent events - Real-world applications of conditional probabilities to problem solving	Exam/Skills Demo

	• The use of random numbers in computing • Methods for generating pseudo- random numbers • The significance of randomness to modern computing systems e.g. PKI (Public Key Infrastructure)	
LO3	Demonstrate knowledge and understanding of basic computer simulation methods to include: • Computational methods • Numerical methods e.g. Monte Carlo methods	Skills demo
LO4	Apply knowledge and understanding of numeric and structural representations and usages to address computational approaches to real-world problems • e.g.: practically employ use cases, of arrays/matrices e.g. in graphics, games scenarios, in Google PageRank algorithm, sequence alignment or nearest neighbour problems	Skills demo
LO5	Apply knowledge and understanding of Know-how and skill range Demonstrate a broad range of specialised skills and algorithms to address computational approaches to real-world problems • e.g.: assess the appropriateness of an algorithm or computational approach, for example in terms of speed, efficiency, best/expected/worst case behaviours	Exam
LO6	Apply knowledge and understanding of elementary probability and information-theory to address computational approaches to real-world problems • e.g.: Conduct average case analysis of algorithms, perform failure prediction, perform software branch prediction, analyse network packet loss	Skills demo/Exam
LO7	Differentiate between the principles of modelling and simulation and explain their use of abstraction that allows a machine to address a real-world problem • e.g. modelling the temperature in server room, or robot control, or software for an autonomous vehicle	Skills demo

LO8.	Identify approaches to problem definition, solution design, testing and evaluation	Skills demo
LO9.	Outline the strengths and weaknesses of, and primary areas of application for, a range of contemporary problem definition and analysis techniques e.g. brute force, divide-and-conquer and heuristic strategies	Exam
LO10	Distinguish between pragmatic solving of a problem (using logic to treat the symptoms) and semantics (interpretation of the problem to establish the root cause)	Exam/Skills Demo
LO11	Demonstrate awareness of the role of personal attributes such as initiative, methodical approach, logical reasoning, persistence and lateral thinking, in prevention and resolution of problems	Skills demo
LO12	Engage in an iterative process involving model creation and validation of the correspondence between the model and the real-world situations being modelled e.g.: create a simple, formal mathematical model of a real-world situation and use that model in a simulation e.g. modelling of the temperature in server room, or robot control, or software for an autonomous vehicle, or modelling traffic flow at an intersection	Skills demo
LO13	Reflect on the impact of employing numerical and logical thinking principles and concepts in the real world, including accuracy, precision, and decisions that may be made based on computer simulations and models	Skills demo/Exam

11c. Guidelines for Assessment Activities

The assessor is required to devise assessment briefs for the Skills Demonstration and examination papers, marking schemes and outline solutions for the Examination. In devising the assessment briefs and examination papers), care should be taken to ensure that the learner is given the opportunity to show evidence of achievement of ALL the learning outcomes. Assessment briefs may be designed to allow the learner to make use of a wide range of media in presenting assessment evidence, as appropriate. Quality assured procedures must be in place to ensure the reliability of learner evidence

Skills Demonstrations	70%
The skills demonstration may be carried out as a series of individual or combined tasks as outlined below. The overall length of the Skills Demonstrations is 20 hours	
Table 11 indicates that 7 of the Learning Outcomes are being assessed by both an examination and a skills demonstration. This is intended to give the assessor flexibility when devising assessment briefs and an examination paper. The assessor must ensure that it is a different aspect of the LO that is assessed on each occasion, guard against double assessment and ensure that all learning outcomes are fairly and equally assessed. Skills Demonstration 1- 10% LO1, LO13 - The assessor will devise a practical skills assessment that allows the user to demonstrate their achievement and understanding of numeric and structural data with accurate and correct calculations and formula - The assessor will devise a practical salient features of iterative and recursive algorithms and of where to apply each Skills Demonstration 2-15% LO3 and LO4 - The assessor will devise a practical skills assessment that allows the user to demonstrate their achievement and understanding of Computational Modelling and Simulation and Algebra and Applications Skills Demonstration 3-15% LO2 and LO6	

- The assessor will devise a practical skills assessment that allows the user to demonstrate their achievement and understanding of Discrete Probability and Applications of Discrete Probability

Skills Demonstration 4-30% LO7, LO8, LO10, LO11 & LO12

- The assessor will devise a practical skills assessment that allows the user to demonstrate their achievement and understanding of Computational Modelling and Simulation by creating a mathematical model of a real - world situation and using that model in a simulation

Evidence for each Skills Demonstration may take the form of written, oral, graphic, audio, visual or digital evidence, or any combination of these. Any audio, video or digital evidence must be provided in a suitable format.

All instructions for the learner must be clearly outlined in an assessment brief.

Examination	30% LOs 1, 2, 5, 6, 9, 10, 13
This examination should be held at the conclusion of the course, following a suitable period of independent study for learners. The examination is 1.5 hours long and will take place at the end when all the LO's have been delivered.	
The Learner should provide evidence that demonstrates the following: Section A of this examination will consist of 10 short questions, from which the learner will be expected to answer all questions. (10 marks). Section B of this examination will consist of two long questions, from which the learner will be expected to answer all questions (20 marks). Evidence for this assessment technique should be in written form.	

12. Grading

Distinction:	80% - 100%
Merit:	65% - 79%
Pass:	50% - 64%
Unsuccessful:	0% - 49%

Computational Methods and Problem Solving 5N0554	Learner Marking Sheet 1 Skills Demonstrations 70%
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Learner's Name: _____

Assessment Criteria	Max Mark	Learner Mark
Skills Demonstration 1 - Demonstrate knowledge and understanding of numeric and structural data - Accurate calculations and correct use of formula - Coherent format and logical progression of thought	4 4 2	
Subtotal	10	
Skills Demonstration 2 - Demonstrate knowledge and understanding of Computational Modelling and Simulation - Demonstrate knowledge and understanding of linear Algebra and Applications - Accurate calculations and correct use of formula - Coherent format and logical progression of thought	6 6 4 4	
Subtotal	20	
Skills Demonstration 3 - Demonstrate knowledge and understanding of Discrete Probability - Demonstrate knowledge and understanding of Applications of Discrete Probability - Accurate calculations and correct use of formula - Coherent format and logical progression of thought	6 6 4 4	
Subtotal	20	
Skills Demonstration 4 - Demonstrate knowledge and understanding of Computational Modelling and Simulation - Create mathematical model of a real - world situation and use that model in a simulation - Accurate calculations and correct use of formula - Coherent format and logical progression of thought	6 6 4 4	
Subtotal	20	
Total Mark	70	

Assessor's Signature: _____

Date: _____

External Authenticator's Signature: _____

Date: _____

Computational Methods and Problem Solving 5N0554	Learner Marking Sheet 2 Examination 30%
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Learner's Name: _____

Assessment Criteria	Max Mark	Learner Mark
Section A: 10 short questions, answer all 10 (1 mark each) Question No.:* _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ 1 mark 1 mark 1 mark 1 mark 1 mark 1 mark 1 mark 1 mark 1 mark 1 mark 1 mark		
Subtotal	10 marks	
Section B: (Answer all questions) 2 Structured long questions (10 marks each) Question No.:* _____ _____	10 marks 10 marks	
Subtotal	20 marks	
Total Mark	30 marks	
Total mark	30%	

Assessor's Signature: _____

Date: _____

External Authenticator's Signature: _____

Date: _____